

Audience-Driven Data A Persona-Based

NBA Player Efficiency Analysis (2014-

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How the same findings require different approaches for audience

10 Final

14

11

14

1 What This Project Is

The project was to analyze NBA player data and create one that demonstrates my thinking I did that and completed. But somewhere between the initial feedback and the final deliverable, I got curious about a different question that was part of the original project.

The early feedback was helpful—focus on distributions first, then look at bivariate relationships and trends over time. Pretty standard data science workflow. But one line stood out: *"I wish I could help more, but I'm not very familiar with basketball."*

That got me thinking. A comment from my professor—someone who clearly understands data science—felt like he couldn't engage fully with the content because he doesn't follow basketball. How many other people would look at the same infographic and just...tune out? Not that the analysis is bad, but because the way it's presented assumes a certain baseline knowledge of basketball that some people just don't have.

So I started asking myself: what if I took the exact same dataset and created two different presentations for completely different types of people? Someone who lives on r/NBA and someone who's never watched a basketball game. A business executive who understands ROI versus a college student just getting into data science. Would the design choices need to be different, or could I just "simplify" one version and call it a day?

That question turned into a parallel exploration that ran alongside the main project. I figured out the design choices that worked best for each audience. The exploratory work helped me figure out what my primary audience should prioritize, but it also became interesting on its own.

2 The Core

If I have one analytical insight—let's say "player efficiency matters more than scoring"—how would I communicate that to four completely different audiences?

- Someone who follows advanced NBA stats
- A business executive who watches sports casually but doesn't know basketball metrics
- A college student curious about data viz but with zero basketball knowledge
- Someone scrolling social media who doesn't care about sports at all

The standard advice is "make it simpler for beginners" or "add more details for experts." But that treats audience adaptation like a volume knob—just turn the complexity up or down. I suspected it was more than that. Different audiences might need different color schemes, different narrative hooks, different graph types, even different emotional appeals.

I wanted to test whether that was actually true.

3 How I Approached

3.1 Building Audience

I started by mapping out who these different people actually are—not just demographics like age or education, but how they think, what motivates them, how they consume information.

I pulled from a few sources: Pew Research studies on sports consumption, Nielsen reports on digital behavior, and some academic papers on graph literacy (turns out about 40% of people struggle with basic charts, which was eye-opening).

Here's what I ended up with:

Persona Analytics

- The people who live on Basketball Reference, follow FiveThirtyEight's RAPTOR argue about PER Win Shares on
- They want methodological They'll actually read your They about outliers and contrarian
- Jargon isn't a barrier for them—it's a signal that the content is

Persona Sports-Aware

- Business professionals who watch March Madness or follow the NFL, but don't know "usage rate" means in
- They understand efficiency quadrant frameworks—concepts that across sports and business
- They need the analysis framed in language that doesn't require NBA

Persona Curious

- People exploring data visualization for the first time, maybe for a class or personal
- They need the concept explained before the jargon feels like gatekeeping to
- Visual metaphors work better than scatter They want a story, not a statistical

Persona Unaware/

- People who encounter content passively while scrolling—zero sports interest, zero background
- They need a provocative hook on a theme recognition value) not basketball
- One big number, minimal text, scroll-stopping
- I didn't build this More on that

3.2 The Data Work (Standard)

This part was the same across all I used the NBA player-season dataset from Reference

- 2,156 player-seasons across 648 unique
- 10 completed seasons (2014-15 through 2023-
- Filtered for players with at least 20 minutes per game (the regulars, not bench

The analytical findings were consistent no matter who the audience

1. Centers have the highest average PER (19.2) despite being the least common
2. Playoff players average 10.6% higher PER than non-playoff
3. PPG and PER correlate strongly ($r = 0.759$), but there are outliers like Jokic who the
4. Scoring increased 16.6% over the

Exce handled most this— AVERAGEIF CORREL PivotTables The chanics weren't

3.3 Using AI to Explore Narrative

Here's where things got more complex. After I had the numbers, I needed to figure out how to frame them for each audience.

I used NotebookLM for this. I'd upload the data outputs along with the persona and prompt it with something like, "Given this efficiency dataset and an audience of professionals who don't follow basketball, what framing would make them interested?"

What came back wasn't a final product—it was perspective options I wouldn't have thought of on my own.

- For generalists, it suggested an ROI salary contribution, efficiency gaps, fairness.
- For newcomers, it proposed the "invisible star" metaphor—players who don't score but make their teammates better.
- For enthusiasts, it pointed to Jokic as the key to low usage, disproportionately impactful.

I'd then iterate on those ideas, cross-check against my Excel calcs to make sure the data held up and design the visuals. The AI wasn't making the infographics—it was helping me think through what story each audience would find compelling.

3.4 Documenting Every Design Decision

For each variant, I documented what I chose and why, grounded in actual research rather than just gut feeling.

Example: Why did I use orange for the "scoring specialist" in the newcomer version and yellow-green for the "efficiency star"? Because research on color psychology shows warm colors (like orange and red) evoke dominance and aggression, while yellow-green evokes cooperation and growth. That's not a random aesthetic choice—it's a semantic one. The colors are chosen to categorize before the text even starts.

I ended up with 30+ documented decisions across the three variants—everything from font density to graph types to where I placed visual elements.

3.5 Project Development

This exploration happened alongside the primary project, with each feeding into the other.

February: After topic selection and dataset analysis, I established the structure distributions, then bivariate trends. This became the backbone of every early feedback loop.

March: While refining the core research on graph sports fandom, I built the persona profiles—pulling in and business communication. The goal was understanding not just *what* to show, but *how* different audiences would process it.

April-May 2026: Used NotebookLM to test narrative angles for each persona. Same completely different results. This revealed that audience adaptation isn't about simplifying or complexifying one design—it's about fundamentally different choices at every stage.

Final synthesis: The primary deliverable incorporated insights from three personas, optimized for sports-aware audiences. The persona exploration directly shaped the final narrative.

4 The Three Variants I

4.1 Variant "The Invisible Star" (For

Goal Explain the concept of efficiency to someone who's never thought about basketball before

Narrative "Winning games isn't just about scoring points—it's about making your mates Some players are 'invisible stars' who don't show up on highlight reels but elevate everyone around

4.1. Design

Color High-Contrast Orange Yellow-

I went with orange for the scorer (30 PTS) and yellow-green for the efficiency player (PTS). Research shows warm colors signal dominance and individual performance, while tones evoke For someone unfamiliar with basketball, those color associations some of the work before they even read the

I avoided muted tones Neutra colors would "this is complex analysis," which would scare off

Visual Stars and

Stars are universally understood as "this person is excellent." No basketball knowledge quired The silhouettes keep it abstract—this isn't about specific players, it's about That lowers cognitive load (concept from multimedia learning research—people process shapes faster than detailed

I specifically avoided scatter plots or bar Studies show about 40% of adults with basic graph If you're trying to welcome someone into data don't lead with the format that might intimidate

The "+20%"

One big efficiency players increase their shooting by That' the whole Heath & Heath's "Made to Stick" framework talks about this—unexpected, concrete messages are A table with five metrics

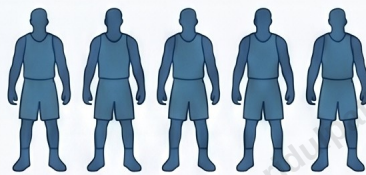
Zero

I used "efficiency," "multiplier," "invisible impact"—all plain For newcomers, nical jargon reads as They interpret it as "this content isn't for me." I wanted opposite

The Invisible Star: Beyond the Scoreboard

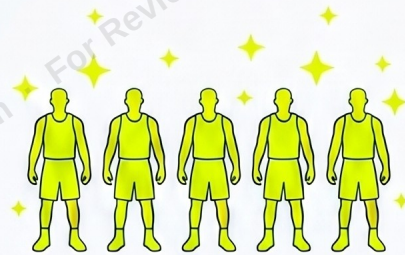
Winning games requires more than just high scoring
—it requires high-efficiency teammates.

The Scoring Specialist: A Solo Act



Teammates' shooting
percentage stays the
same or drops.

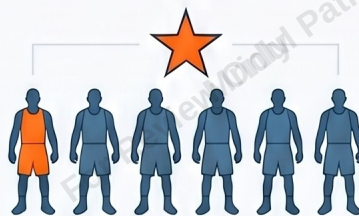
The Efficiency Star: The Great Multiplier



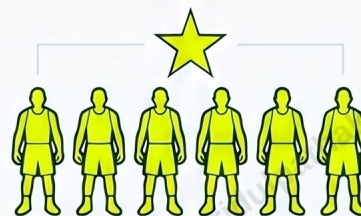
+20%

Increases teammates'
scoring chances by 20%.

The "Invisible" Impact on Team Success



SOLO ACT TEAM:
Harder to Win



MULTIPLIER TEAM:
Win More Often

Teams with "Multipliers" win more often because
the whole unit becomes harder to defend.



NotebookLM

Figure Variant "The Invisible Star" — Designed for people with zero basketball knowledge. High-contrast colors (orange, yellow, blue) and metaphoric visuals (stars, silhouettes) make the concept memorable. The goal was instant understanding.

4.2 Variant "ROI on the Court" (For

Goal Make the analysis accessible to business-minded people who follow sports but not specific

Narrative "Which players deliver the best value for their salary? Let's look at this an investment

4.2. Design

Color Tan and Earth

The tan brown and navy dots—this was I wanted it to like financial document, not a sports entertainment. Neutral earth tones signal and analytical. It's the visual equivalent of the Wall Street Journal

I avoided bright team colors (Lakers purple, Celtics green) because that would make it tribal. Generalists aren't invested in specific teams—they want objective

Scatter Plot with Quadrant

X-axis is salary, Y-axis is Win Shares (contribution to team). Reference lines split into high cost/low output, high cost/high output, low cost/low output, low output

This structure is familiar from the BCG Growth-Share Matrix, which most business grads have. Quadrant thinking is intuitive for this audience—they immediately "top right is

I skipped regression lines and confidence. Too for people who stats-

Faint Basketball Court

There's a subtle court shape in the. It's not prominent enough to basketball expertise but it's enough to activate what's called a "schema"—brain sees the court and goes "okay, this is sports-related, I can engage with

Business Terminology, Not NBA

"ROI," "salary efficiency gap," "gold standard anchors," "league average"—these are terms that transfer across sports and business. I avoided PER, RAPTOR, usage. Those are insider

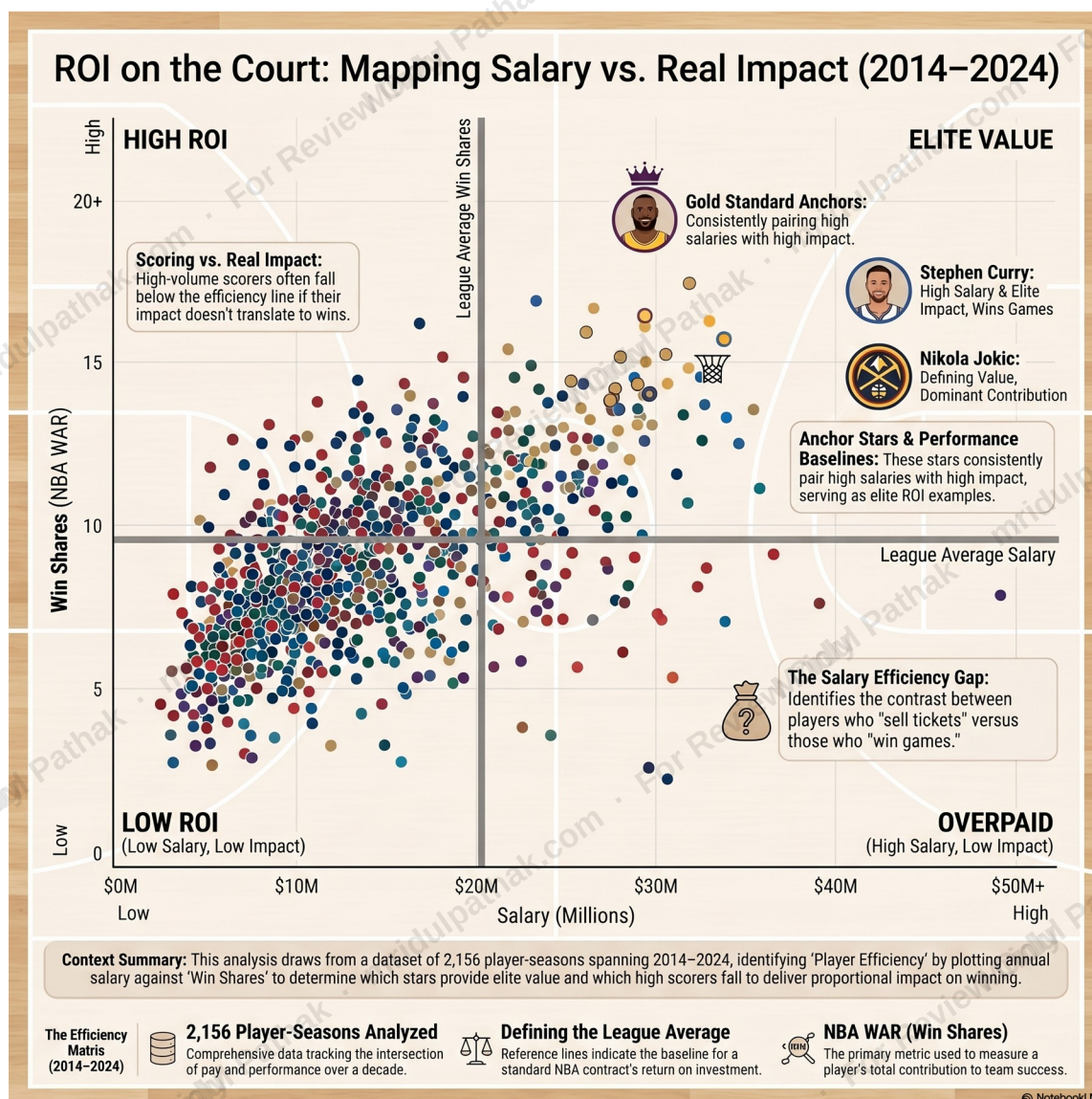


Figure Variant "ROI on the Court" — Designed for business Muted tones, scatter plot with quadrant framework (familiar from BCG matrix), faint court outline context, business-framed Bridges sports analytics and business

4.3 Variant "Efficiency Volume" (For

Goal: Give analytics nerds something they can't get from a generic sports site—rigor and contrarian

Narrative "Does the efficiency-volume trade-off actually hold? Let's look at the who break the

4.3. Design

Color Gradient Encoding RAPTOR

The dots aren't just random colors—they encode a third variable (RAPTOR Impact Coo tones purple) represent low warm tones gold) represent impact This follows ColorBrewer guidelines for sequential Enthusiasts expect data The gradient adds a dimension without cluttering chart

Scatter Plot + Regression Line + Confidence

This tests a well-known hypothesis in analytics as usage efficient typically declines (you can't maintain peak performance when you're taking 30 shots a

The dashed regression line shows the expected The shaded confidence shows statistical Outliers above the line—like Jokic—are the interesting

I kept the confidence interval because this audience audits If you don't uncertainty, they'll question your

Player-Specific

Jokic (2020- Low usage, disproportionately high RAPTOR. This is the famous volume paradox

Curry (2015- 30% usage, still 5-8% above adjusted efficiency The 73-9 season where he broke the

Enthusiasts care about *who* breaks the pattern, not just that the pattern

Advanced Metrics and Methodology

"ERA-Adjusted True Shooting %," "RAPTOR Impact Score," "Regression "Sample Size Requirements"—this is the language this audience cargon validates the content is for If I wrote in plain English, they'd dismiss it as pop

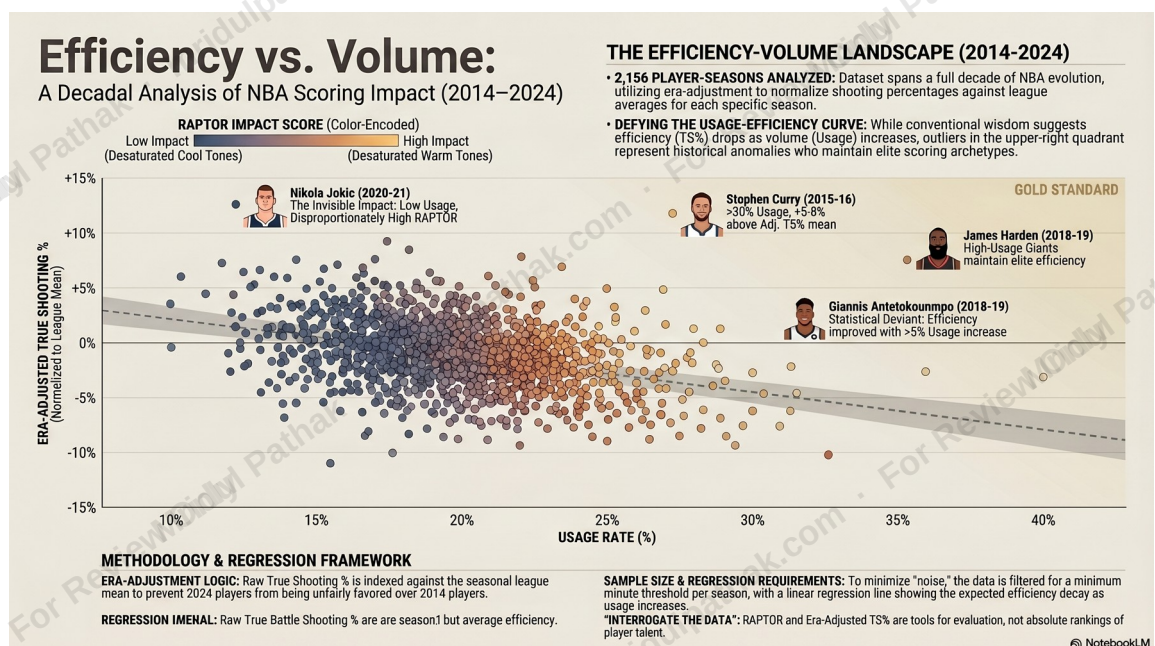


Figure Variant "Efficiency Volume" — Designed for analytics Sequential color gradient (encoding RAPTOR), scatter plot with regression + confidence interval, player callouts (Jokic, Curry, Harden, Giannis), advanced metrics, methodology Prioritizes statistical rigor and contrarian

5 Why I Didn't Build the Fourth

The fourth persona—people with zero sports interest who encounter content while scrolling require a completely different validation

Based on research, it would

- Near-monochrome design with one neon accent (for scroll-stopping)
- One massive "hero number" and fewer than 50 words
- A provocative, non-sports headline like "Is This America's Most Underpaid

- Universal human themes (fairness, value, recognition) instead of basketball

But here's the creating scroll-stopping content for disinterested audiences metrics I don't have access Click-through rates on different headline Dwell time minimal detailed Sharing

The other three variants are grounded in actual audience research—surveys on sports graph literacy business communication The fourth would be I'd rather skip it than fake

6 Primary Generalist-Optimized

Based on the persona exploration, I optimized the primary deliverable for sports-aware alists—people comfortable with analytical thinking but unfamiliar with domain-specific **Design**

- **Navy background with structured bar charts:** Professional aesthetic, not focuse
- **Plain-English metric definitions** "PPG Points Per Game — avg scored per game per season." No assumed
- **Structured sections following analytical workflow:** Univariate distributions → poral trends → Bivariate relationships → Performance
- **Moderate jargon with** Uses terms like "Pearson $r = 0.759$ " but explains as "a strong positive
- **Data transparency** Sample size ($n =$ filter criteria (20 source citation all

How It Synthesized All Three

From the newcomer Plain-English definitions, narrative section headers, tional framing that doesn't assume basketball

From the generalist Professionacolor business-appropriate chart (bar charts, scatter plots, tables), moderate jargon with

From the enthusiast Data transparency (methodology notes, sample size), player callouts (Jokic, Embiid, Curry), correlation coefficients

Technical

The visual tools demonstrate standard analytical

- Bar COUNTIF, AVERAGEIF for categorical
- Line Time-series analysis with seasonal
- Scatter CORREL for bivariate
- Data Conditional formatting and

The design balanced accessibility narrative hooks) with rigo (transparency, statistic validation) That hybrid approach came directly from testing different audiences engage with the same

7 What I Learned from

7.1 Color Isn't Just

Different audiences read color palettes as signals before they process the actual

- High-contrast primaries (orange, yellow, blue) = "this will be easy to
- Muted earth tones (tan, brown, olive) = "this is professional
- Sequential gradients (cool to warm) = "this has statistical

If there's a mismatch—like colors paired with dense regression analysis—it cognitive The design is sending mixed

7.2 Jargon Works Like an Audience

Technical terminology doesn't just convey It signals who the content is

For experts, jargon validates that you respect their For novices, it reads as isn't for you." You can't solve that by adding a glossary to a single You need designs

7.3 The Same Data Tells Different

My NBA dataset supports all of these narratives

1. "Invisible stars make teammates better" (newcomer
2. "Salary efficiency reveals market gaps" (generalist
3. "Jokic defies the efficiency-volume trade-off" (enthusiast

Effective communication isn't about finding *the* story in *that's* about choosing story that resonates with *this specific*

7.4 AI as an Iteration

The value of NotebookLM wasn't producing final It was compressing weeks of storming into

It suggested the ROI framing for generalists when I was stuck thinking in terms of PER. proposed the "invisible impact" metaphor for It identified Jokic as the key for

Best use perspective not content I'd iterate with the validate and build the visuals

8 Where This Framework Applies Beyond

This isn't just about sports The persona-driven approach works anywhere you to communicate the same information to different

8.1 Product Strategy Internship Application (Summer

This I'm working as a Product Strategy intern at a marketing infrastructure in Part of the role involves creating dashboards and reports for two distinct types using the same platform

- **Content** These users care about engagement metrics—views, watch time, dience For them, dashboards need simplified visualizations with takeaways ("Your best posting time is 7 PM on They want not raw
- **Local businesses:** These users care about conversion metrics—leads generated, co acquisitionROI on marketing They need business-framed competitiv benchmarking, and budget efficiency

Same backend data But the frontend presentation—what metrics first, what language describes them, what level of visual complexity matches the user's tication—needs to adapt based on who's logging

The persona framework from this project is directly informing how I'm approaching work It's not anymore—it's how I'm deciding which dashboard modules to for which user

8.2 Research

Same study, three

- Academic Full methodology, regression tables, p-
- Industry Executive summary, case examples, actionable
- Public Plain-English summary, visual metaphors, real-world

8.3 MAT: Idea Validation Platform (In

I'm building a research aggregation that helps people validate business The challenge is that the same market intelligence needs to be presented differently depending where someone is in their ideation

- **Early exploration phase:** Users at this stage are exploring whether an idea has at They need high-level market sizing, trend snapshots, and comparable case The is to build confidence and reduce friction—too much overwhelms paralyzes decision-
- **Validation** Users here are stress-testing the idea against They competitive landscape breakdowns, customer segmentation data, and financial frameworksThis is where detailed analysis actually helps—they're ready for because they're trying to make an informed go/no-go
- **Pitch** Users at this stage are packaging the idea for investors or holders They need curated statistics formatted for slide visua charts for presentations, and risk/opportunity matrices that tell a persuasive

Same research API serving all three But the UI adapts complexity, detail depth, narrative tone based on detected user intent—essentially applying this project's persona-visualization framework to information

The question I'm working through how do you detect intent without making the user out a survey? Right now I'm testing signals (how quickly they're whether they're downloading just what questions they're asking) to infer which they're in and adapt the presentation

9 Limitations and What I'd Do

9.1 No User

I built the personas from secondary research—Pew Nielsen academic on graph I didn't test these variants with actual If I were doing this properly, I'd A/B test with persona-matched audiences and

- Comprehension (can they accurately summarize the finding after
- Engagement (how long do they spend with
- Trust (do they find it
- Recall (what do they remember a week

9.2 Single

I only tested this on sports Color psychology jargon visua literac patterns might work differently for healthcare metrics, financial data, environmental Generalization is an open

9.3 AI Tool

NotebookLM's design outputs limited the stylistic A future version would compare generated variants against what professional designers would create for the same Do they make similar audience-driven choices? Different

1 Final

This started as a practical what should my primary deliverable prioritize? The feedback suggested focusing on then bivariate then Tha gave me a structural

But that comment — "I'm not very familiar with sports" — made me realize the structure the only The same structure could work for multiple audiences if the design choices jargon, visual complexity, narrative framing—adapted to who was looking at

The exploration became interesting beyond just "how should I complete this project." turned into a framework for thinking about how visual communication changes based on ence

The primary deliverable incorporated insights from three But the themselves revealed something effective data visualization isn't about finding perfect It's about understanding who you're designing for and what they need visualization to do for

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